Data Cleaning & Reporting Automation Using Python

Overview

This project demonstrates an automated workflow for data cleaning and reporting using Python. The task focuses on handling missing values, duplicate records, inconsistent text formatting, and generating visual summaries from a raw dataset.[page:1]

Objective

The purpose of this project is to understand how data preprocessing improves data quality and how automation makes reporting faster and more efficient. The workflow is designed so the same notebook can be reused for similar raw datasets in future tasks.[page:1]

Dataset Description

The raw dataset contains sales-related records with columns such as OrderID, Date, Customer, Region, Sales, and Category. It intentionally includes common data quality issues such as missing values, duplicate rows, inconsistent capitalization, and inconsistent date formats so that the cleaning process can be demonstrated clearly.[page:1]

Data Cleaning Process

The data cleaning workflow was automated in Python using pandas. Duplicate rows were removed, missing values were handled, text fields were standardized, dates were converted into a consistent datetime format, and numeric sales values were cleaned for further analysis.[page:1]

Reporting Automation

After cleaning the dataset, summary tables and visual reports were generated automatically. These included total sales by region, total sales by category, and a daily sales trend chart, which help present the cleaned data in a more understandable form.[page:1]

Outcome

The final output of the project includes a cleaned dataset, summary insights, and charts that demonstrate reporting efficiency. This module helps build understanding of preprocessing, workflow automation, and practical reporting using Python.[page:1]